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19

Lecture 19

Memory Management I: Hierarchy and Contiguous Allocation

by Brijesh Shukla

C07: Operating Systems

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Today’s Agenda

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- Introduction to Memory Management
- Memory Hierarchy and Characteristics
- Contiguous Allocation: Fixed Partitions
- Dynamic Partitioning Algorithms
- Fragmentation and Solutions
- Swapping and Overlays

Introduction to Memory Management

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🧠 Definition

Memory management is how the **OS controls and optimizes main memory** for running processes.

⚙️ Core Responsibilities

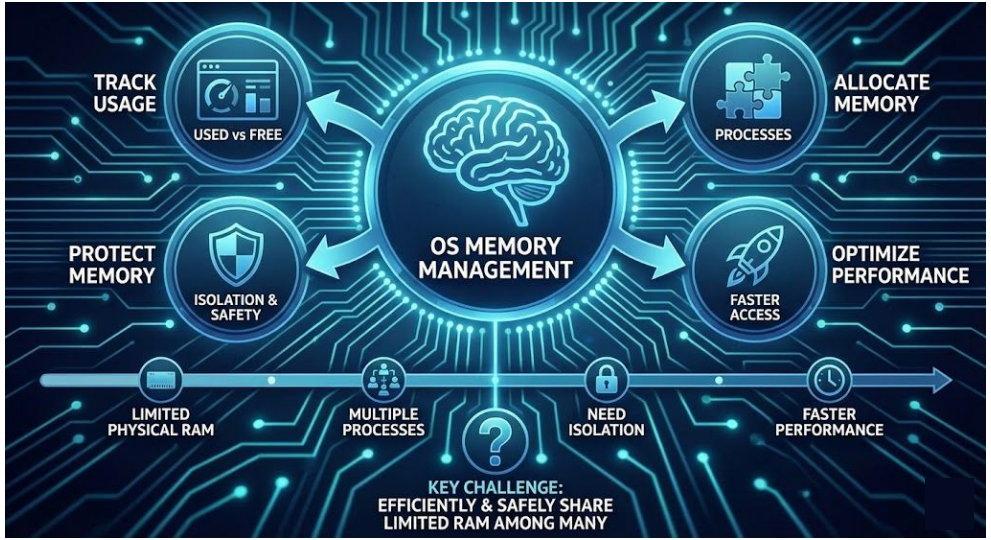
- 📊 Track **used vs free** memory
- ➕ Allocate memory to processes
- ➖ Free memory when done
- 🔐 Protect process memory
- 🚀 Optimize performance

❓ Why It Matters

- 🧱 Physical memory is **limited**
- 👥 Multiple processes compete
- 🔒 Need isolation & safety
- ⏱️ Faster memory = better performance
- ♻️ Efficient use of available RAM

🎯 Key Challenge

How can limited physical memory be shared **efficiently and safely** among many processes?



Objectives of Memory Management

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- 🔄 **Relocation**
  Load programs **anywhere in memory**; support dynamic execution
- 🔐 **Protection**
  Isolate processes; **protect OS/kernel** memory
- 🤝 **Sharing**
  Controlled memory sharing (libraries, IPC)
- 🧩 **Logical Organization**
  Support modules, separate compile & link
- 💾 **Physical Organization**
  Move data between RAM ↔ storage, **transparent to user**
- ⚡ **Efficiency**
  Minimize waste, faster access, better CPU use

Key Takeaway
Memory management balances **safety, flexibility, and performance** while using limited RAM efficiently.

The Memory Hierarchy

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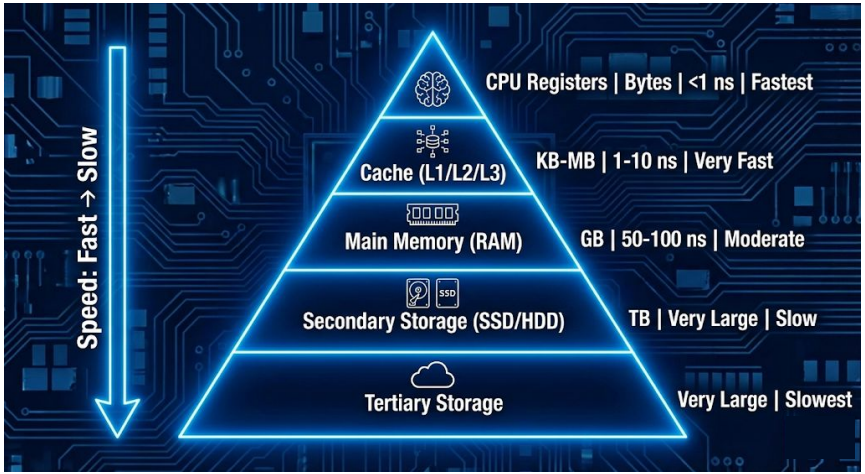
🧠 **Core Idea**
Memory is organized in **layers** based on **speed, cost, and capacity**
➡️ **Faster = Smaller & Costlier**
➡️ **Slower = Larger & Cheaper**

🏎️ **Hierarchy (Top → Bottom)**

- ⚡ **CPU Registers**
  Bytes | **< 1 ns** | Fastest, most expensive
- 🚀 **Cache (L1 / L2 / L3)**
  KB–MB | **1–10 ns** | Very fast
- 💾 **Main Memory (RAM)**
  GB | **50–100 ns** | Moderate speed & cost
- 📀 **Secondary Storage (SSD/HDD)**
  TB | **µs–ms** | Slow, cheap
- ☁️ **Tertiary Storage (Tape/Cloud)**
  Very large | **seconds+** | Slowest, cheapest



🎯 **Key Takeaway**
Systems trade **speed for size and cost**—fast memory is small, slow memory is large.

Memory Hierarchy Characteristics

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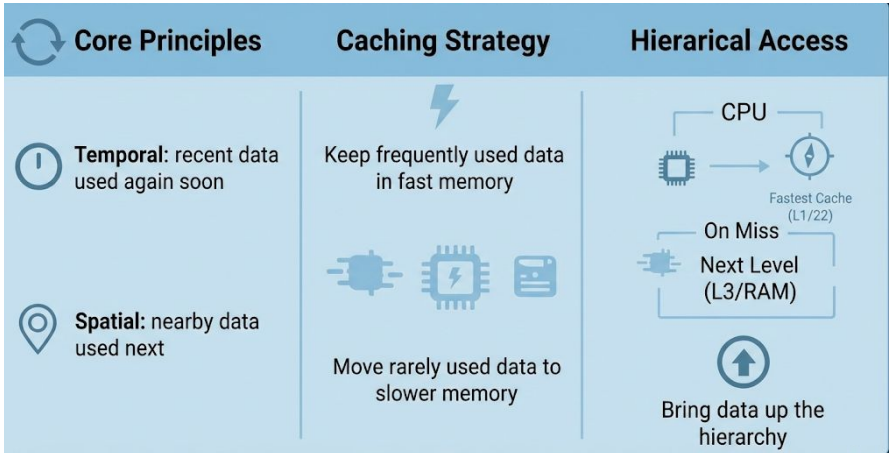
📌 Core Principles

- 🔁 **Locality of Reference**
  - **Temporal:** recent data used again soon
  - **Spatial:** nearby data used next
- ⚡ **Caching Strategy**
  - Keep **frequently used** data in fast memory
  - Move **rarely used** data to slower memory
- 🧭 **Hierarchical Access**
  - CPU checks **fastest level first**
  - On miss → check next level
  - Bring data **up the hierarchy**



📊 Performance Impact

- **Hit Rate:** % found in fast memory
- **Effective Access Time:** `Hit Rate × Fast Access + Miss Rate × Slow Access`

Key Takeaway

High locality + good caching = faster memory access and better performance.

Main Memory Organization

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🧠 Memory Basics

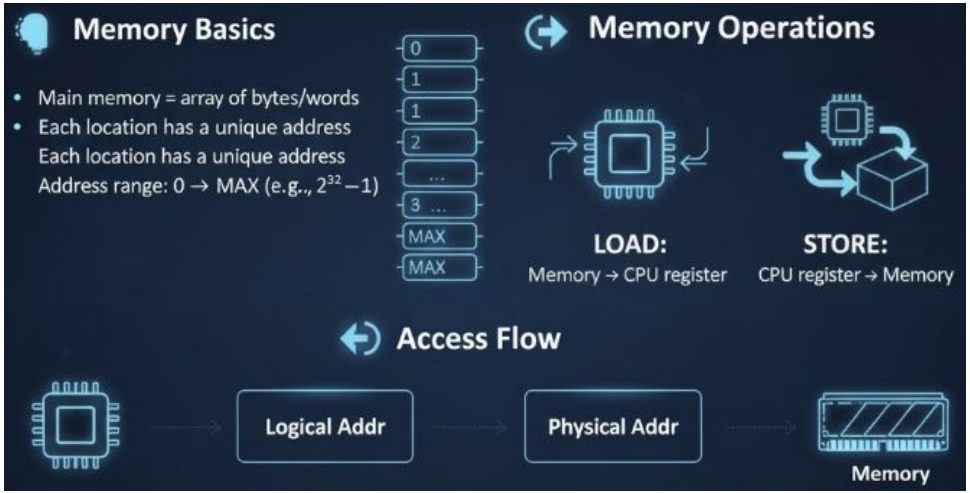
- Main memory = array of bytes/words
- Each location has a **unique address**
- Address range: `0 → MAX` (e.g., $2^{32} - 1$)

🔄 Memory Operations

- **LOAD:** Memory → CPU register
- **STORE:** CPU register → Memory

🧭 Address Spaces

- **Physical Address**
  - Real RAM address
- **Logical / Virtual Address**
  - Generated by CPU/process
  - Translated to physical address



🔁 Access Flow

`CPU → Logical Addr → Translation → Physical Addr → Memory`

Main Memory Organization

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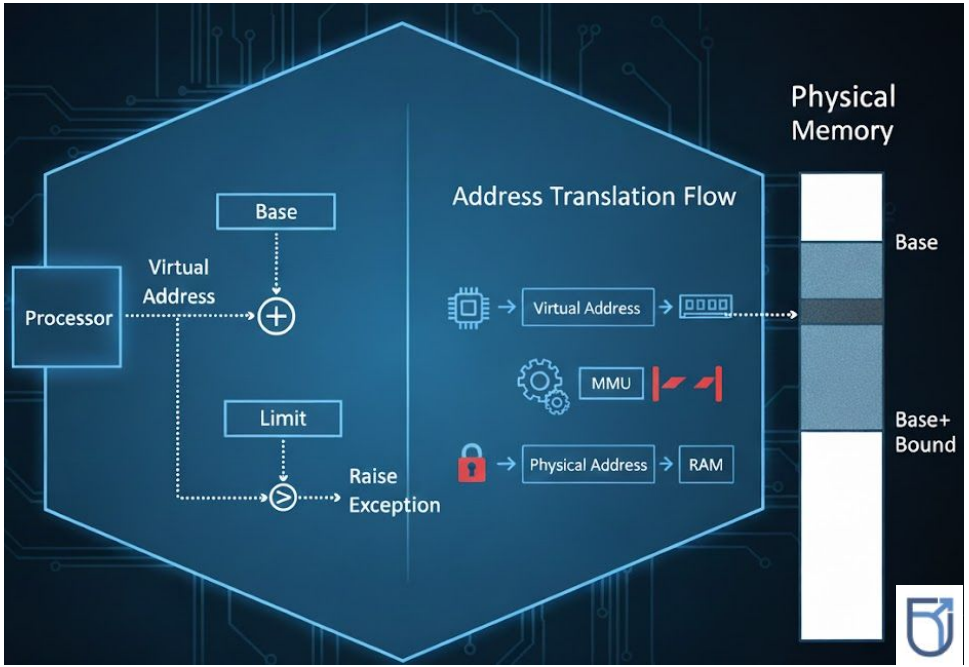
🔐 **Memory Protection**

- **Base Register:** start of process memory
- **Limit Register:** size of process memory
- **Check:** `Base ≤ Address < Base + Limit`

Key Takeaway

Programs use virtual addresses; hardware safely maps them to physical memory.



Address Binding

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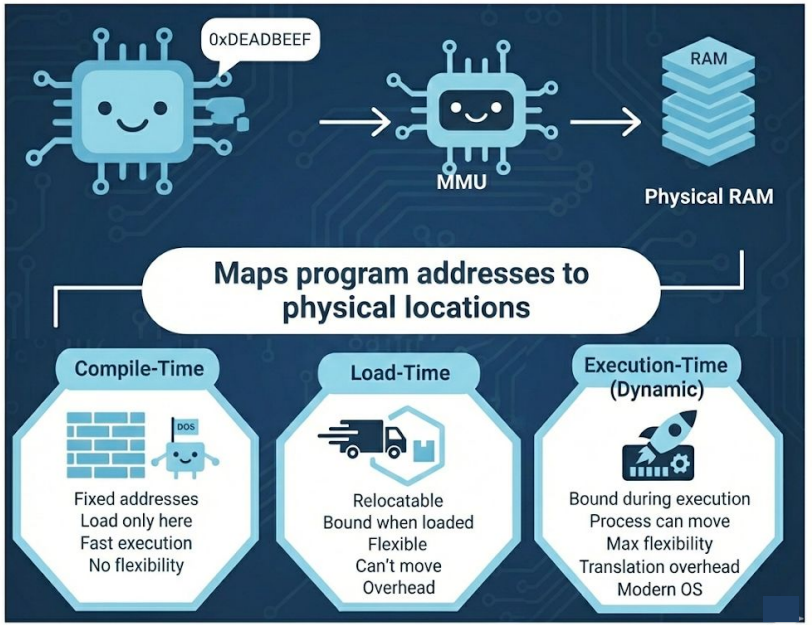
🧩 **What is Address Binding?**
Mapping **program addresses** to **actual memory locations**

🕑 **Types of Binding**

- 🧱 **Compile-Time**
  - Fixed (absolute) addresses
  - Load at one location only
  - ✔ Fast | ❌ No flexibility
  - Example: old DOS programs
- 📦 **Load-Time**
  - Relocatable code
  - Bound when loaded
  - ✔ More flexible | ❌ Can't move while running
- 🔄 **Execution-Time (Dynamic)**
  - Bound during execution
  - Process can move in memory
  - ✔ Maximum flexibility | ❌ Translation overhead
  - Used in modern OS



Address Binding

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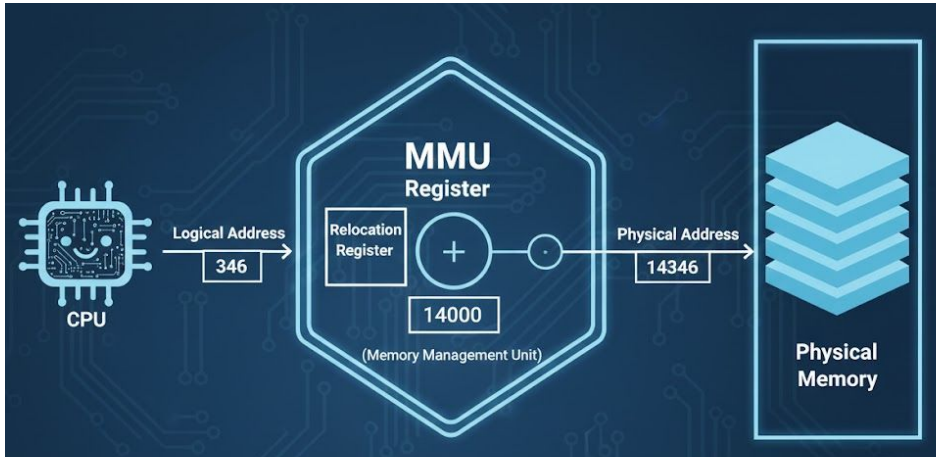
⚙️ **Address Translation**
`Logical Address + Relocation Register → Physical Address`
Handled by **MMU (hardware)**

Key Takeaway
Later binding = more flexibility; modern systems use **execution-time binding** for safety and efficiency.



Protection in Contiguous Allocation

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🔐 **Why Protection Is Needed**

- Prevent process → process memory access
- Protect OS from user programs
- Catch invalid memory references
- Ensure stability & security

⚙️ **Hardware Support: Base & Limit Registers**

- **Base Register:** Start address of process
- **Limit Register:** Size of process memory

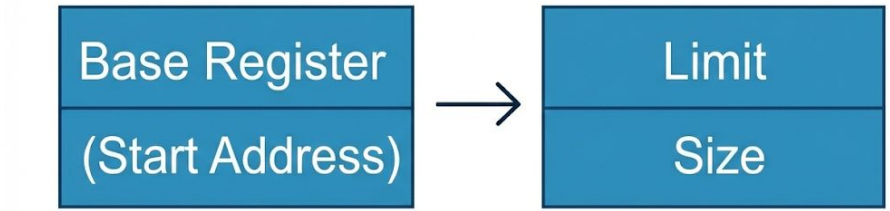
Check on every access:

```
Physical Address = Base + A
```

- If `A < 0` or `A ≥ Limit` → **TRAP**
- Else → **Access allowed**

📊 **Example**

- Base = **300KB**, Limit = **120KB**
- Valid range: **300–420KB**



Physical Address = Base + A

If A < 0 or A ≥ Limit → TRAP

Else → Access allowed

Protection in Contiguous Allocation

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🧠 **Where It's Used**

- **Fixed Partitions:**
  - Base = partition start
  - Limit = partition size
- **Dynamic Partitions:**
  - Base & limit set per allocation

🚨 **What Protection Catches**

- Buffer overflows
- Invalid pointers
- Out-of-bounds array access

➡️ OS raises **exception** and terminates process

🧑‍💻 **Privilege Levels**

- **Kernel Mode:**
  - Access all memory
  - Can change base/limit
- **User Mode:**
  - Restricted to its memory only

Key Takeaway

Base & limit registers provide **simple, fast, hardware-level memory protection** in contiguous allocation systems.

Contiguous Memory Allocation

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🧠 **Definition**

Each process occupies **one continuous block** of physical memory.

📦 **Core Idea**

- Process uses **consecutive addresses**
- No gaps **inside** a process
- Entire process loaded as **one block**

🗂️ **Simple Memory Layout**

- OS in resident memory
- User processes placed **one after another**
- Remaining space = free memory

🔀 **Two Approaches**

- 🧱 **Fixed Partitioning**
  Predefined, fixed-size partitions
- 📐 **Dynamic Partitioning**
  Partitions created based on process size

✅ **Advantages**

- Simple design, Fast access and Low overhead

❌ **Disadvantages**

- Fragmentation, Poor flexibility and Limits process size

Key Takeaway

Contiguous allocation is fast and simple, but wastes memory and lacks flexibility.

Fixed Partitioning - Concept

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🧠 **Definition**
Memory is split into **fixed-size partitions** at system startup.
➡️ **One process per partition**

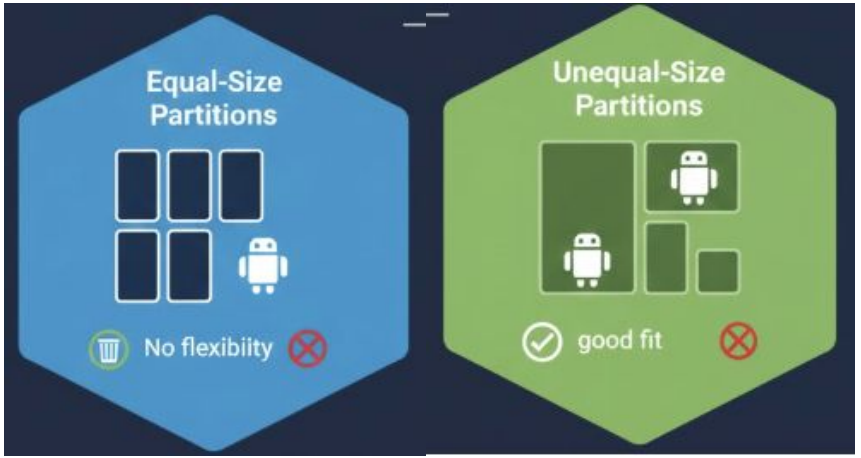
⚙️ **Key Characteristics**

- Partition sizes decided **before boot**
- Sizes **do not change** during execution
- Process must **fit entirely** in a partition

🔀 **Variants**

- 🟦 **Equal-Size Partitions**
  Same size; simple; causes more waste

- 🟩 **Unequal-Size Partitions**
  Different sizes; better fit for varied processes



Fixed Partitioning - Concept

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📦 **Example (Equal Size: 5 MB)**

- P1 (3 MB) → **2 MB wasted**
- P2 (1 MB) → **4 MB wasted**
- P3 (4 MB) → **1 MB wasted**

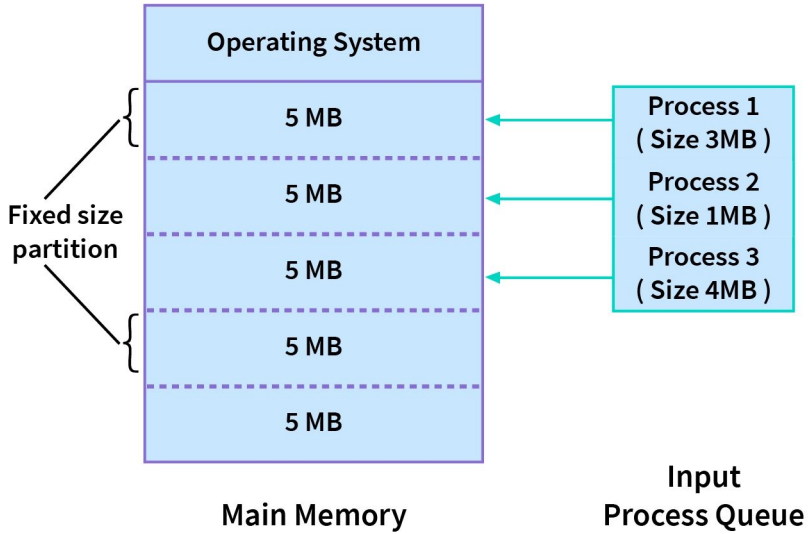
➡️ **Internal Fragmentation**

✅ **Advantages**

- Simple & predictable
- Easy allocation

❌ **Disadvantages**

- Internal fragmentation
- Fixed process limit
- Poor fit for large/variable processes



Key Takeaway

Fixed partitioning is easy to manage but wastes memory due to internal fragmentation.

Fixed Partitioning – Placement Strategies

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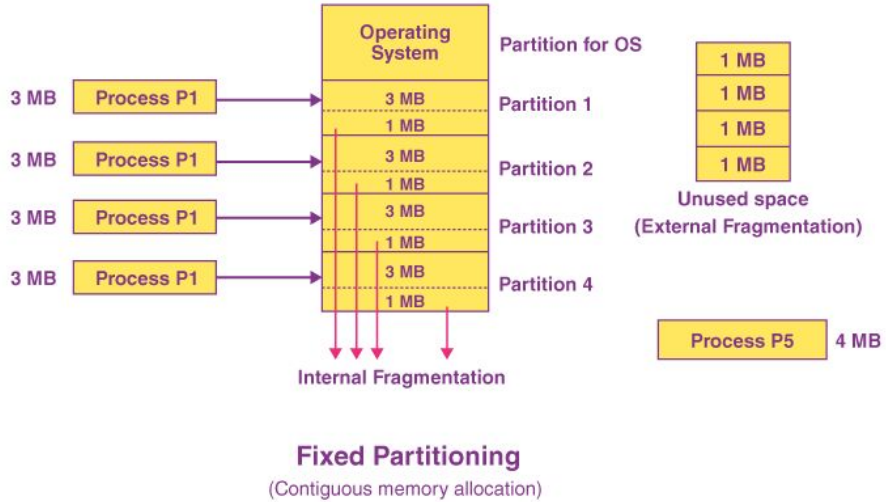
✅ **Best Practices**

- Prefer **unequal partitions**
- Match sizes to workload
- Review & adjust over time

Key Takeaway

Placement choice trades **flexibility** for **simplicity**. Internal fragmentation is unavoidable in fixed partitioning.



Fixed Partitioning

(Contiguous memory allocation)

Dynamic Partitioning – Concept

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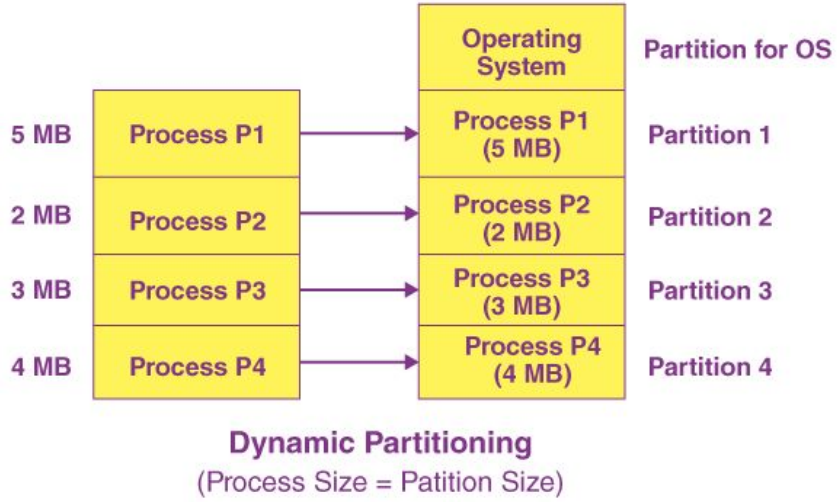
🧠 **Definition**
Memory partitions are **created on demand**, sized **exactly to the process**.

⚙️ **Key Features**

- No fixed partition sizes
- **Partition = process size**
- Created on load, removed on exit
- ❌ No **internal fragmentation**

📊 **How Memory Changes**

- Start: OS + large free block
- Load processes → memory splits dynamically
- Free space shrinks and shifts over time



Dynamic Partitioning
(Process Size = Patition Size)

Dynamic Partitioning – Concept

✅ **Advantages**

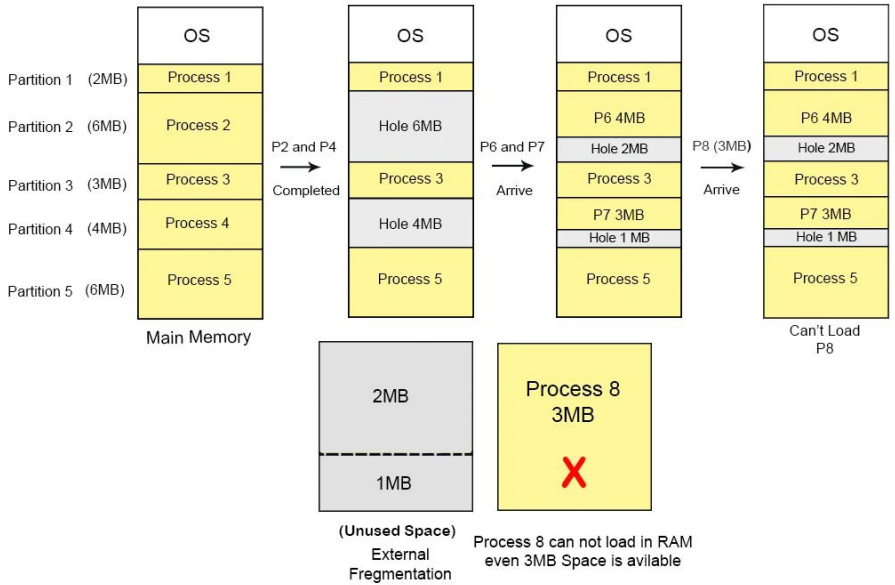
- No internal waste
- Flexible for varied process sizes
- Better initial memory use

❌ **Disadvantages**

- **External fragmentation** develops
- More complex allocation logic
- Overhead tracking free blocks

Key Takeaway

Dynamic partitioning fits processes perfectly but suffers from external fragmentation over time.



First Fit Allocation Algorithm

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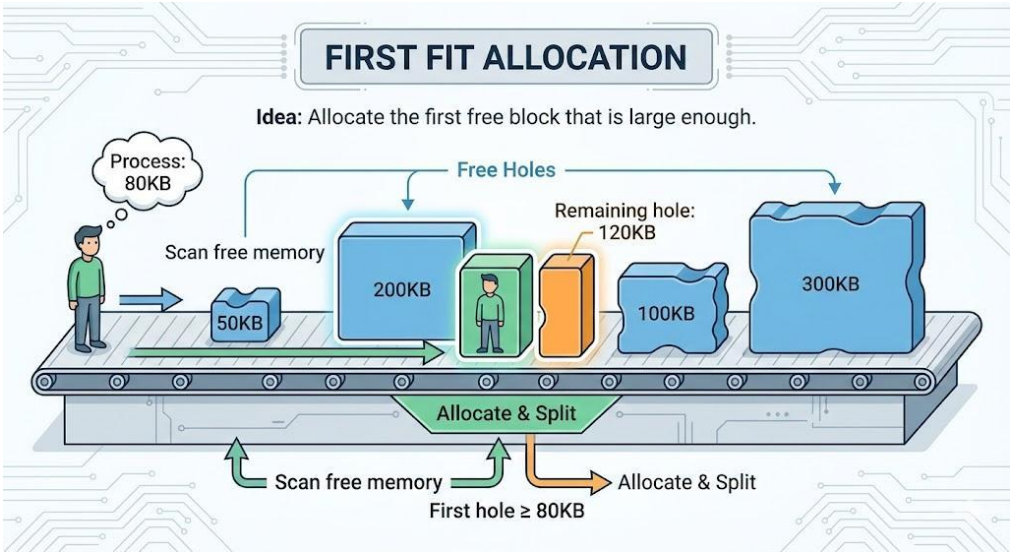
🧠 **Idea**
Allocate the **first free block** that is **large enough** for the process.

⚙️ **How It Works**
1️⃣ Scan free memory **from the start**
2️⃣ Pick the **first hole ≥ process size**
3️⃣ Allocate memory
4️⃣ Split hole if needed
5️⃣ Stop searching

📌 **Example**
Free Holes: 50KB | 200KB | 100KB | 300KB
Process: 80KB
➡️ **Chosen:** 200KB (first fit)
➡️ **Remaining hole:** 120KB



First Fit Allocation Algorithm

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✅ **Pros**

- Fast & simple
- Low overhead
- Works well in practice

❌ **Cons**

- Small holes accumulate at start
- Not always optimal
- Fragmentation near beginning

⏱️ **Performance**

- **Time:** `O(n)` (scan free list)

Key Takeaway

First Fit is quick and simple, but may cause fragmentation over time.

First Fit Allocation Algorithm

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Block / Size	Available	Can Occupy P3?	Memory Wastage After Process Occupies
Block 1 (20K)	Yes	No	
Block 2 (100K)	No		
Block 3 (40K)	No		
Block 4 (200K)	No		
Block 5 (10K)	Yes	No	

None of the available blocks
Can accommodate P4. It remains unallocated

Practice Problem 1

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A system uses **fixed partition memory management** with the following partitions: **100KB, 500KB, 200KB, 300KB, 600KB**

The following processes arrive in order: **P1 = 212KB, P2 = 417KB, P3 = 112KB, P4 = 426KB.**

Allocate the processes using the **First Fit** algorithm. Show clearly which partition each process is allocated to. Identify any process that cannot be allocated. Calculate the **total internal fragmentation**.

Allocation (First Fit):

- **P1 (212KB)** → 500KB partition → waste = 288KB
- **P2 (417KB)** → 600KB partition → waste = 183KB
- **P3 (112KB)** → 200KB partition → waste = 88KB
- **P4 (426KB)** → ❌ no partition fits → **waiting**

Answers:

- **Allocated:** P1, P2, P3
- **Waiting:** P4 (1 process)
- **Total Internal Fragmentation:** 288 + 183 + 88 = **559KB**

Explanation: In fixed partitioning, process must fit entirely in one partition. Unused space inside partitions causes **internal fragmentation**.

Best Fit Allocation Algorithm

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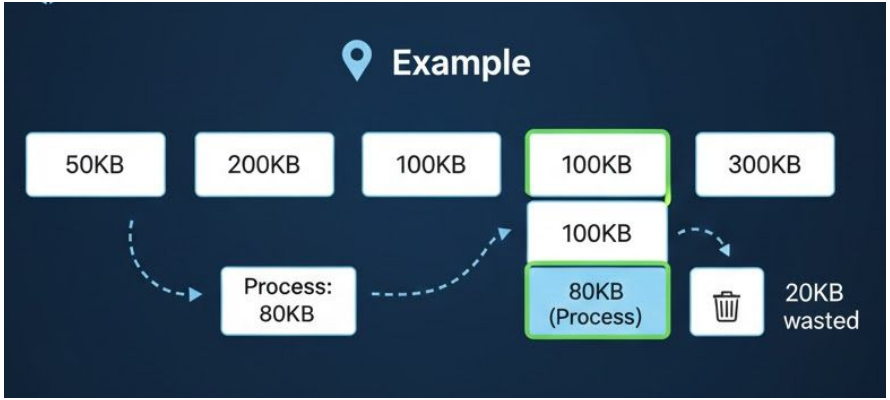
🧠 **Idea**
Allocate the **smallest free block** that can fit the process
➡️ Leaves **minimum leftover space**

⚙️ **How It Works**
1️⃣ Scan **entire free list**
2️⃣ Choose **smallest hole ≥ process size**
3️⃣ Allocate & split remainder

📌 **Example**
Free Holes: 50KB | 200KB | 100KB | 300KB
Process: 80KB
➡️ **Chosen:** 100KB (waste = 20KB)



Best Fit Allocation Algorithm

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✅ **Pros**

- Minimizes waste per allocation
- Saves large holes for big processes
- Good utilization (in theory)

❌ **Cons**

- Slower (full scan)
- Creates many tiny holes
- Often worse than First Fit in practice

⏱️ **Performance**

- **Time:** `O(n)` (checks all holes)

Key Takeaway

Best Fit reduces immediate waste but increases fragmentation and overhead over time.

Best Fit Allocation Algorithm

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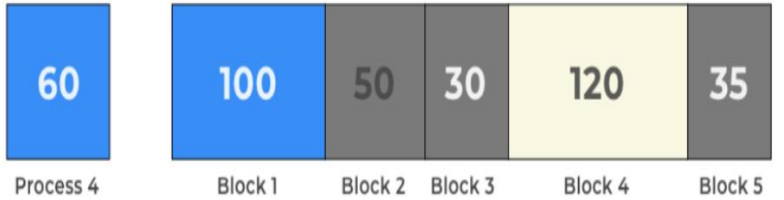

Block / Size	Available	Can Occupy P3?	Memory Wastage After Process Occupies	
Block 1 (100K)	Yes	Yes	100 - 60 = 40	Best
Block 2 (50K)	No	-	-	
Block 3 (30K)	Yes	-	-	
Block 4 (120K)	Yes	Yes	120 - 60 = 60	
Block 5 (35K)	Yes	-	-	

Block 1 results in the least memory wastage

P4 goes to Block 1

Worst Fit Allocation Algorithm

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🧠 **Idea**

Allocate the **largest available hole**

➡️ Leave a **big leftover** that might still be useful

⚙️ **How It Works**

1️⃣ Scan **entire free list**

2️⃣ Pick **largest hole ≥ process size**

3️⃣ Allocate & keep large remainder

📌 **Example**

Free Holes: 15KB | 25KB | 40KB | 50KB

Process: 30KB

➡️ **Chosen:** 100KB

➡️ **Remaining:** 70KB

Worst Fit Allocation Algorithm

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✅ **Pros**

- Leftover space stays large
- Fewer tiny holes (initially)

❌ **Cons**

- Slow (full scan)
- Breaks large holes quickly
- Often **worst performer**

⏱️ **Performance**

- **Time:** `O(n)`

Key Takeaway

Worst Fit tries to preserve usable space, but usually causes poor overall memory utilization.

Worst Fit Allocation Algorithm

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	Available	Can Occupy P4?	Memory Wastage After Process Occupies
Block 1	No		
Block 2	No		
Block 3	Yes	No	P4 remains unallocated, As can not be fit by any Free blocks
Block 4	No		
Block 5	Yes	No	

Comparison of Allocation Algorithms

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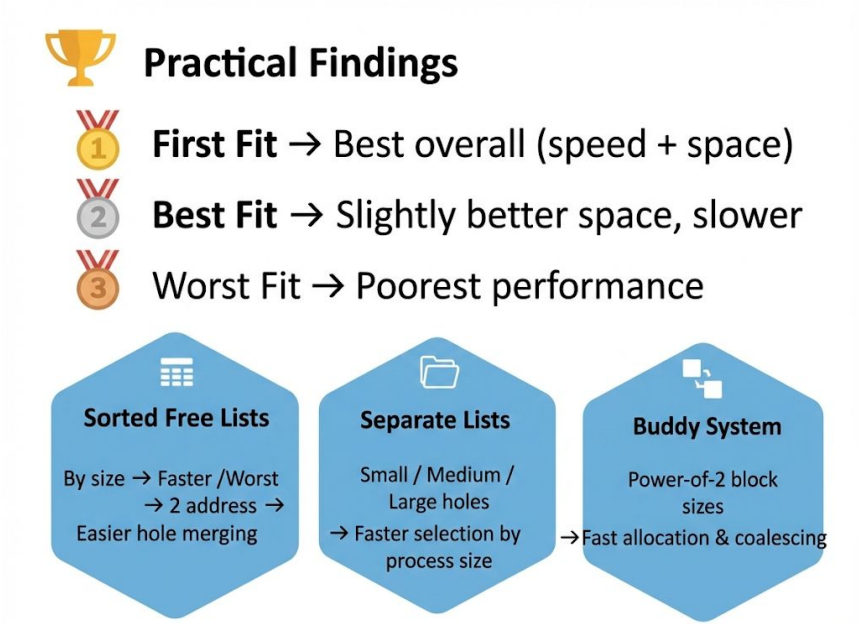
Practical Findings

- 🥇 **First Fit** → Best overall (speed + space)
- 🥈 **Best Fit** → Slightly better space, slower
- 🥉 **Worst Fit** → Poorest performance



✅ **Recommendation**

First Fit is preferred in practice: **simple, fast, and efficient**.

Practice Problem 2

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A system uses **fixed partition memory management** with the following partitions: **100KB | 200KB | 50KB | 300KB**

The processes arrive in the following order: **P1 = 70KB, P2 = 110KB, P3 = 40KB**

1. Allocate the processes using: (a) **First Fit** (b) **Best Fit** (c) **Worst Fit**
2. Show clearly which partition each process is allocated to under each algorithm.
3. Identify any unallocated process (if any).
4. Calculate the **total internal fragmentation** for each algorithm.

✅ **Best Fit**

- P1 (70) → 100 → **30KB internal**
- P2 (110) → 200 → **90KB internal**
- P3 (40) → 50 → **10KB internal**

Total Internal Fragmentation = 130KB
Unused partition: 300KB

✅ **First Fit**

- P1 (70) → 100 → **30KB internal**
- P2 (110) → 200 → **90KB internal**
- P3 (40) → 50 → **10KB internal**

Total Internal Fragmentation = 30 + 90 + 10 = 130KB
Unused partition: 300KB

✅ **Worst Fit**

- P1 (70) → 300 → **230KB internal**
- P2 (110) → 200 → **90KB internal**
- P3 (40) → 100 → **60KB internal**

Total Internal Fragmentation = 230 + 90 + 60 = 380KB
Unused partition: 50KB

Explanation:

- **First Fit:** Fast, practical
- **Best Fit:** Less waste per allocation, many tiny holes
- **Worst Fit:** Leaves big holes but performs worst overall

Memory Fragmentation

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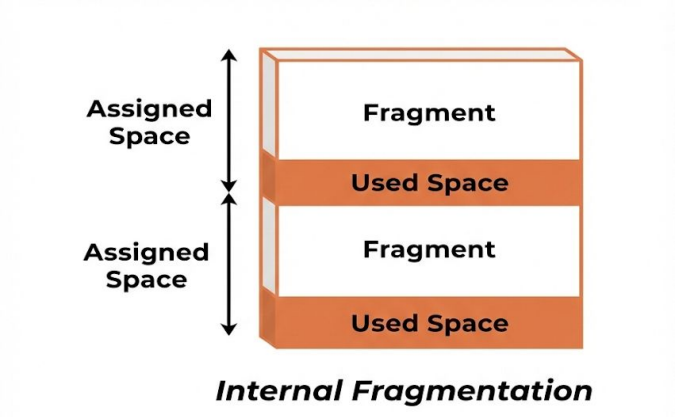
🧠 **Definition**
Fragmentation = **wasted memory** that cannot be effectively used.

🔹 **1. Internal Fragmentation**
- Waste **inside allocated memory**
- Happens in **Fixed Partitioning**
- Allocated > Needed

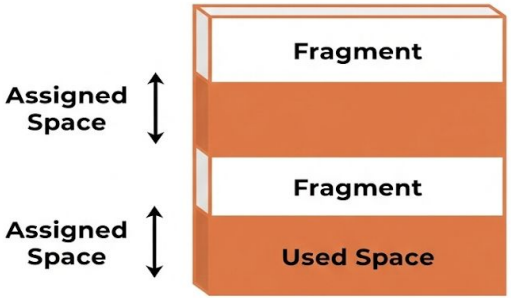
Example:
100KB partition − 70KB process = **30KB wasted**
Formula:
```
Internal = Partition Size - Process Size
```

🔸 **2. External Fragmentation**
- Waste **between processes**
- Happens in **Dynamic Partitioning**
- Free memory exists but **not contiguous**

Key Problem:
Enough total free memory, but **no single large block**



Internal Fragmentation



External Fragmentation

Solutions to Fragmentation – Compaction

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🧠 **What It Is**

Rearrange memory so **all free space becomes one large block**.

🔄 **How It Works**

- Move all processes to one end
- Merge scattered holes
- Update process addresses

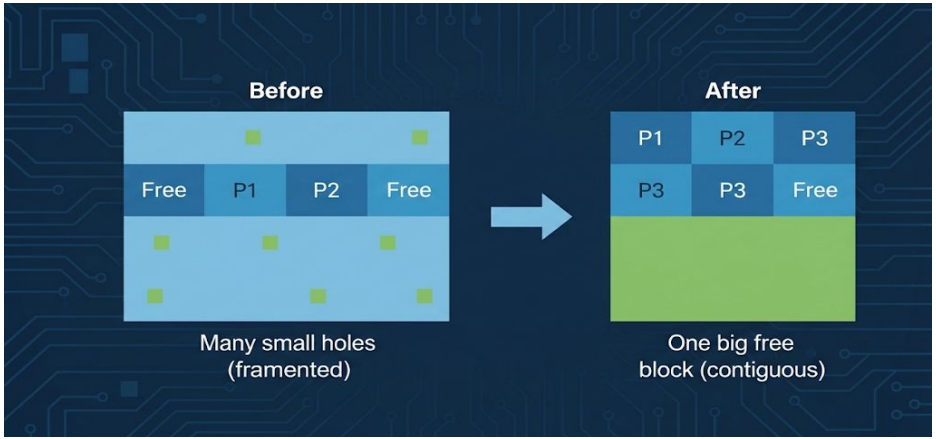
📊 **Effect**

- **Before:** Many small holes (fragmented)
- **After:** One big free block (contiguous)

⚙️ **Requirements**

- Dynamic relocation
- Hardware support (MMU)

Processes movable at runtime



Solutions to Fragmentation – Compaction

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✅ **Pros**

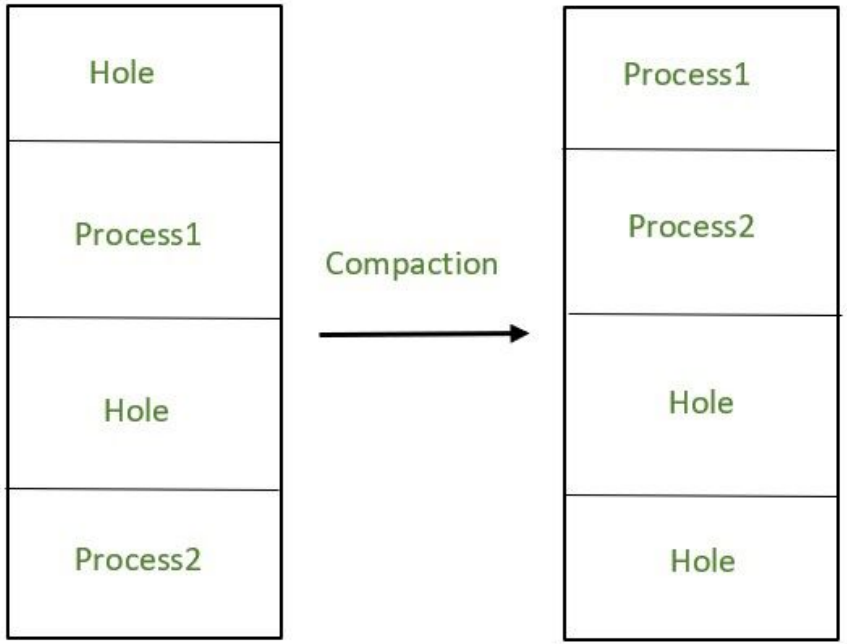
- Eliminates **external fragmentation**
- Improves memory utilization
- Enables large allocations

❌ **Cons**

- **Very expensive** (memory copying)
- System pause required
- Address updates needed
- Time ≈ **O(memory size)**

⏱️ **When to Use**

- Allocation fails despite enough total free memory
- Low system activity
- Fragmentation crosses threshold



Key Takeaway

Compaction fixes fragmentation but is costly—used sparingly in real systems.

Solutions to Fragmentation – Coalescing

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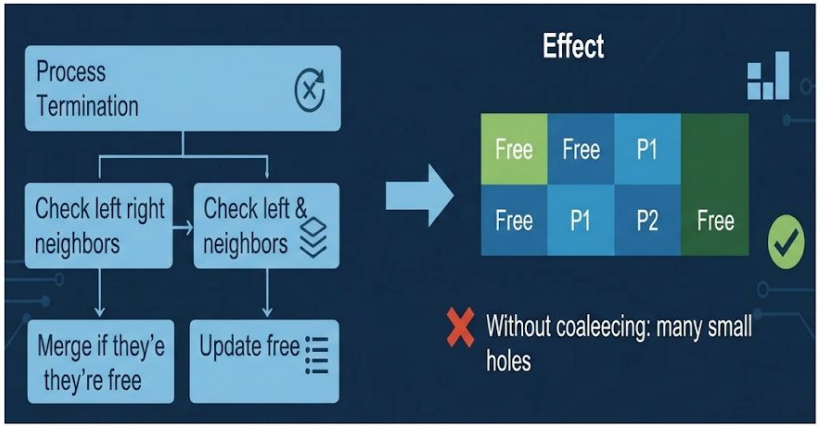
🧠 **What Is Coalescing?**
Automatically **merge adjacent free blocks** when memory is released.

🔄 **How It Works**

- On process termination:
  - Mark block as free
  - Check **left & right neighbors**
  - **Merge** if they're free
- Update free list

📊 **Effect**

- ❌ Without coalescing: many small holes
- ✅ With coalescing: **one larger hole**



Solutions to Fragmentation – Coalescing

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⚙️ **Implementation**

- 🏷️ **Boundary Tags**
  - Size/status at both ends
  - **O(1)** neighbor checks
- 🔗 **Doubly Linked Free List**
  - Easy merge with neighbors

✅ **Pros**

- Prevents fragmentation from worsening
- Fast (**O(1)**)
- No process relocation
- Automatic

❌ **Cons**

- Doesn’t remove existing fragmentation
- Extra bookkeeping

Key Takeaway

Coalescing keeps free memory blocks large by merging neighbors—cheap, fast, and essential for dynamic allocation.

Practice Problem 3

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A system has a total main memory of **1024 MB**. The operating system occupies **124 MB** of memory. Four processes are currently loaded into memory: P1 = 200 MB, P2 = 150 MB, P3 = 100 MB, P4 = 180 MB

After allocation, the remaining free memory is scattered into the following holes: **60 MB, 40 MB, 90 MB, and 80 MB**

```
| OS 124 | P1 200 | Hole 60 | P2 150 | Hole 40 |
| P3 100 | Hole 90 | P4 180 | Hole 80 |
```

1. Calculate the **total memory and free memory.**.
2. Compute the **memory utilization percentage**.
3. Determine the **total external fragmentation**.
4. Identify the **largest available hole**.
5. Determine whether a new process of **120 MB** can be allocated. Justify your answer. Is this possible after compaction?

Calculate Used Memory
Processes: 200+150+100+180=630 MB. Total Used: 124+630=754 MB
Free Memory: 1024−754=270 MB free

🎨 **Scattered Holes**
Holes: 60 MB, 40 MB, 90 MB, 80 MB
Total free: 60+40+90+80=270 MB

📊 **(a) Memory Utilization**
754/1024×100≈73.6%
✔ Utilization ≈ **74%**

📊 **(b) External Fragmentation**
Total Free = **270 MB**
Largest Hole = **90 MB**
Memory is scattered into 4 separate holes → **External Fragmentation exists**

(c) Can a 120MB Process Be Allocated?
Required = 120 MB
Largest hole = 90 MB
Even though total free = 270 MB, ❌ Cannot allocate (no contiguous 120 MB block)

🧩 **After Compaction (if OS Rearranges Memory)**

```
| OS | P1 | P2 | P3 | P4 | Free 270 |
```

Now: ✔ 120MB process can be allocated
Remaining free = 150 MB

Swapping in Memory Management

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🧠 **What is Swapping?**
Moving **entire processes** between **RAM** and **disk** to free memory.

🎯 **Why Swapping?**
- Increase **multiprogramming**
- Free memory for **high-priority** processes

🔄 **How It Works**
- **Swap Out:** RAM → Disk
- **Swap In:** Disk → RAM

⏱️ **Cost**
- Very **expensive** (disk I/O)
- Much slower than context switch

Swap Time
`= Process Size / Disk Transfer Rate`
Example: 100MB ÷ 50MB/s → **2s out + 2s in = 4s**

⚠️ **Key Points**
- Needs backing store (disk)
- Requires contiguous memory (with swapping)
- Excessive swapping → **thrashing**

Key Takeaway
Swapping extends memory capacity but is costly—used carefully in real systems.

Implementing Swapping

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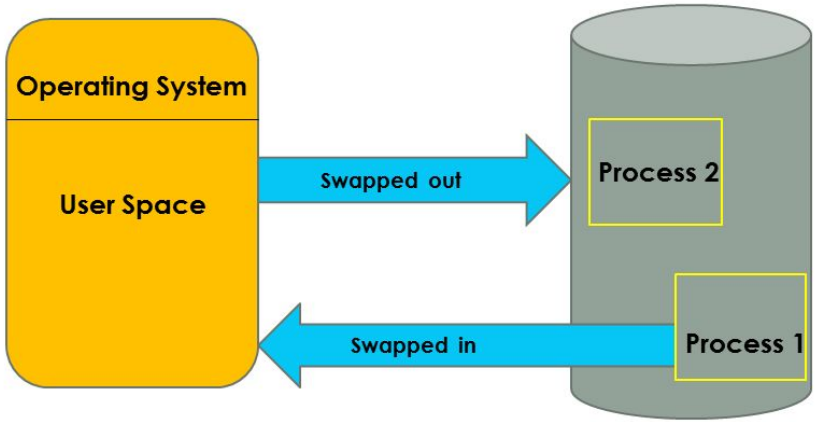
⚙️ **Core Requirements**

- 💽 **Backing Store:** Fast disk space for swapped processes
- 🔁 **Dynamic Relocation:** Process may return to different memory location
- 🔄 **Roll Out / Roll In:**
  - Low priority → swapped out
  - High priority → swapped in



⏱️ **What Affects Swap Time**

- 📦 **Process Size** → larger = slower
- 💾 **Disk Speed** → SSD ≫ HDD
- 🧠 **Memory Load** → may need extra swaps

🧠 **Swapping Policies**

When to Swap Out

- Low free memory, Long I/O wait and Lower priority process

Which Process

- Lowest priority, Longest blocked and Largest memory user

Overlay Technique

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🧠 **What Is It?**

Run programs **larger than RAM** by loading **only needed code/data** at a time.

📜 **Background**

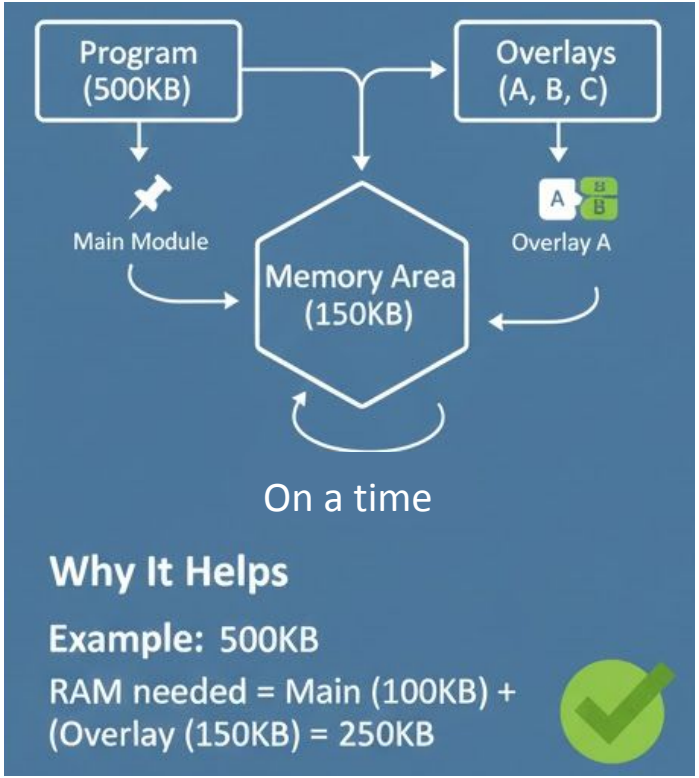
- Used **before virtual memory**
- **Manual** (programmer-managed)
- Common in **early / low-memory systems**

⚙️ **How It Works**

- Program split into **main module + overlays**
- **Main module** stays in memory
- **One overlay at a time** loaded on demand
- Overlays **replace each other** in the same memory area

📊 **Why It Helps**

- Example:
  - Program = **500KB**
  - RAM needed = **Main (100KB) + Largest Overlay (150KB) = 250KB**



Overlay Technique

✅ **Pros**

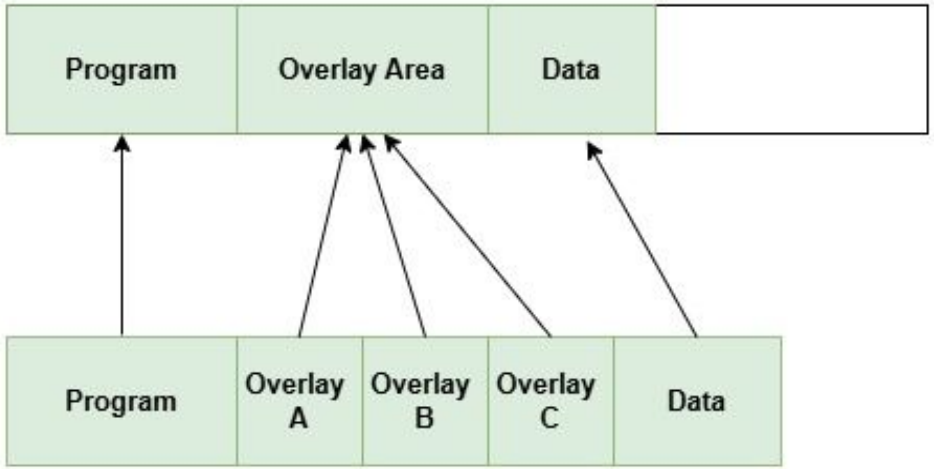
- Run large programs on small memory
- No special hardware
- Predictable memory usage

❌ **Cons**

- Complex & error-prone
- Programmer burden
- Hard to maintain

⚖️ **Overlay vs Virtual Memory**

- **Overlay:** Manual, programmer-controlled
- **Virtual Memory:** Automatic, OS-managed



Key Takeaway

Overlays were an early workaround for limited memory—powerful but painful compared to modern virtual memory.

Partitioning – Pros and Cons

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🧱 Fixed Partitioning

👍 Pros

- Simple & predictable
- Low management overhead
- Easy protection
- No external fragmentation

👎 Cons

- Internal fragmentation (waste)
- Fixed number of processes
- Inflexible sizes
- Large jobs may not fit

🔁 Dynamic Partitioning

👍 Pros

- No internal fragmentation
- Flexible sizes
- Better initial utilization
- Fits varied processes

👎 Cons

- External fragmentation
- Complex allocation
- Compaction is costly
- Variable allocation time

Contiguous Allocation – Pros and Cons

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⚙️ **Overall (Contiguous Allocation)**

👍 **Strengths**

- Fast access
- Simple address translation
- Minimal hardware
- Good locality

👎 **Limitations**

- Fragmentation issues
- Whole process must fit
- Limited sharing
- Size bound by largest hole

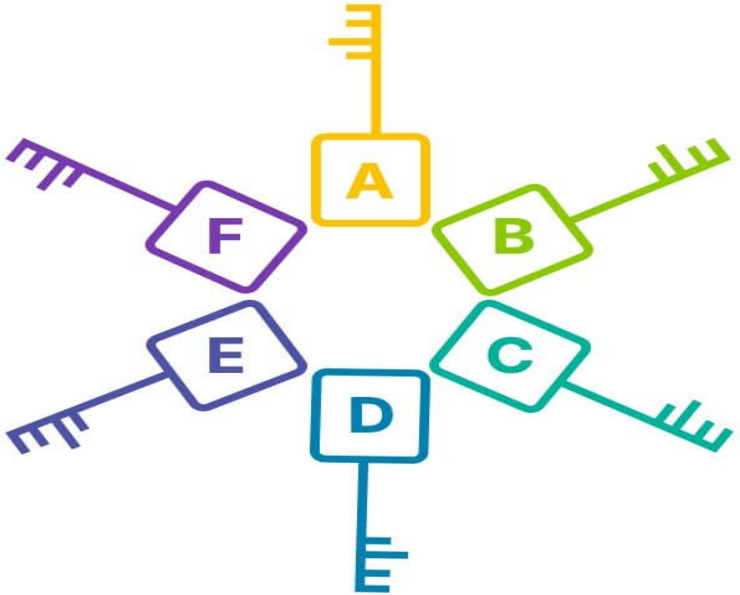
🚀 **Modern Alternatives**

- **Paging:** Removes external fragmentation
- **Segmentation:** Logical program view
- **Virtual Memory:** Beyond physical limits
- **Hybrid:** Paging + Segmentation

Key Takeaway:
Contiguous allocation is fast and simple—but fragmentation limits scalability in modern systems.

Key Takeaways

- Memory Hierarchy
- Contiguous Allocation
- Protection
- Fixed Partitioning
- Dynamic Partitioning
- Allocation Algorithms
- Fragmentation
- Swapping
- Overlays



Thank You!

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Thank you for your attention. Feel free to ask questions or revisit the topics covered.